56. STUDENT GROUP FORMATION WITH SCORE BALANCING

1. AIM

To write a Python program that takes individual student test scores, stores them in a list, and divides the students into equal-sized study groups based on user-specified group size.

2. PROBLEM STATEMENT

Educational institutions divide students into peer learning teams based on batch size constraints. Write a Python program that accepts student scores, validates group size, and partitions student lists into equal sub-groups using list slicing.

3. OBJECTIVES

• Accept student scores into a list scores.
• Read desired group size size and validate against non-positive entries.
• Partition student list into equal slices using range(0, n, size).
• Display group index and score sublists.

4. ASSUMPTIONS & REQUIREMENTS

Assumptions: Student counts and group sizes are positive integers.
Requirements: Python 3.x Interpreter, PyCharm environment.

5. ALGORITHM

Step 1: Initialize list scores = [].
Step 2: Input number of students n.
Step 3: Loop n times to input mark and append to scores.
Step 4: Input size.
Step 5: If size <= 0: display 'Invalid group size'. Else: Loop i from 0 to n in steps of size, slicing scores[i:i + size] and displaying group index.
Step 6: Stop program.

6. PROGRAM FLOWCHART

7. PYTHON PROGRAM (Student Group Formation with Score.py)

```
scores = []

n = int(input("Enter number of students: "))

for i in range(n):
    mark = int(input("Enter student score: "))
    scores.append(mark)

size = int(input("Enter group size: "))

if size <= 0:
    print("Invalid group size")
else:
    group = 1
    for i in range(0, n, size):
        print("Group", group, ":", scores[i:i + size])
        group = group + 1
```

8. EXPLANATION OF THE CODE

• scores: Dynamic list storing student scores.
• size: Defines chunk length for list slicing.
• scores[i:i + size]: Step slicing generating sub-lists for each group.

9. TRACING TABLE

Step	Input Data	List scores	Slicing Range	Group Output
Input Scores	**20, 20, 21, 23, 24**	**[20, 20, 21, 23, 24]**	**-**	**-**
Slice 1	**size = 5**	**[20, 20, 21, 23, 24]**	**0:5**	**Group 1 : [20, 20, 21, 23, 24]**

10. TEST CASES

Test Case	Input	Expected Result	Actual Result
TC 1 (Equal Split)	Students: 5, Scores: 20,20,21,23,24, Group Size: 5	Group 1 : [20, 20, 21, 23, 24]	Group 1 : [20, 20, 21, 23, 24]

11. EXPECTED OUTPUT

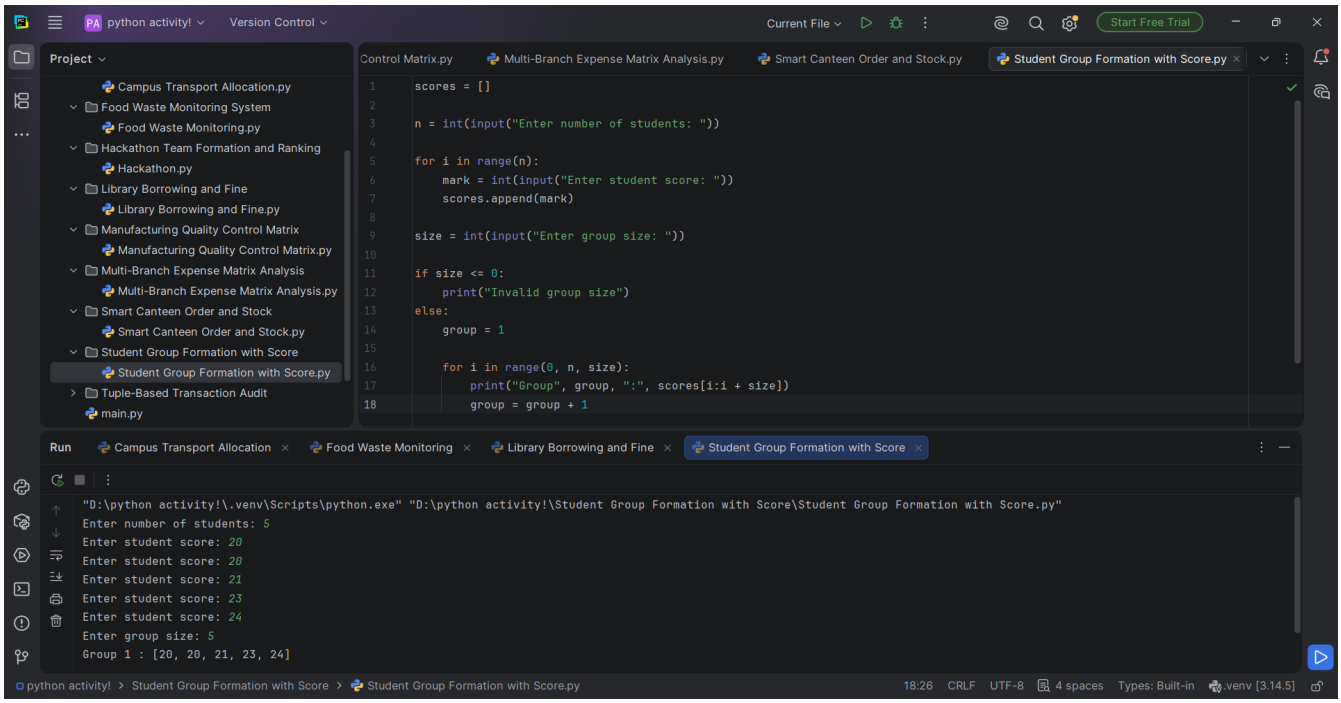
```
Enter number of students: 5
Enter student score: 20
Enter student score: 20
Enter student score: 21
Enter student score: 23
Enter student score: 24
Enter group size: 5
Group 1 : [20, 20, 21, 23, 24]
```

12. ACTUAL OUTPUT

```
Enter number of students: 5
Enter student score: 20
Enter student score: 20
Enter student score: 21
Enter student score: 23
Enter student score: 24
Enter group size: 5
Group 1 : [20, 20, 21, 23, 24]
```

13. PROGRAM EXECUTION SCREENSHOT

Actual Input and Output - PyCharm Run Screenshot



14. RESULT/CONCLUSION

The Python program successfully grouped student scores based on input group sizes using list slicing.